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Mathematical physics made easy(er)

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Abstract

Euclidean volume and some distance equations are proved to be instances of ordered combinations (n-tuples) of elements from a list of abstract, disjoint sets. Volume has a permutation property, where that permutation property is proved to limit a set of unordered elements to a cyclic list of at most 3 elements, e.g., 3 unordered distance dimensions (height, width, and depth) and 3 unordered non-distance dimensions (time, mass, and charge). Where the dimensions of Euclidean distance, time, mass, and charge are each divided into a countably infinite number of same-sized intervals and i in N, the i-th unit interval length of Euclidean distance corresponds 1-1 to the i-th unit interval lengths of time, mass, and charge. Derivations of well-known gravity, charge, electromagnetic, relativity, quantum physics equations and constants from unit ratios are shorter and simpler. The unit ratios also allow simple quantum extensions to classical and relativity equations. The math proofs are verified in Rocq.

[1]

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INTRODUCTION

The current practice in physics is to either define an equation or derive an equation from other equations, for example: 1) Deriving Newton's law of gravity from Kepler's laws[2], or from Gauss's law[3], or from Einstein's field equations[4][5]; 2) Maxwell deriving the electromagnetic wave equations from several empirical laws[3]; 3) Inspired by Gaussian geometry and gravity, Einstein defined the field equations[6]; 5) Inspired by several types of equations, Schrödenger defined a wave equation[7].

The philosophy of reductionism posits that complex phenomena can be reduced to a few simple parts: a few simple objects, relations, operations, etc. But there has been no

reduction of gravity, charge, electromagnetism, relativity, and quantum physics equations into a few simple objects and relations.

And current practice uses many "reasonable" assumptions with no motivation from math proofs. For example, relativity and quantum theory assumes: 1) 3 dimensions of space and 2) the laws of physics are the same at every local Euclidean-like coordinate point[6][5][7][8]. But integration, topology, measure theory, Hilbert spaces, linear algebra, (pseudo-)Riemann geometry, etc. have no restriction on the number of dimensions [9][10][11][5][12]. And although the local coordinate point in a (psuedo-)Riemann space is Euclidean-like[12] the properties of real and complex numbers place no restrictions on functions being the same at every Euclidean-like coordinate[9][10].

The first reduction steps are to provide analytic (set, sequence, limit, combinatorial) proofs that can motivate some common physics assumptions and expose some simple unit interval lengths and simple relations between the unit interval lengths from which many physics equations and constants can be derived. Each proof, in this article, has a corresponding formal proof, which has been verified using using the Rocq proof verification system [13], a verification tool used by mathematicians world-wide. The formal proofs are in the Rocq files, "euclidrelations.v" and "threed.v," which are included as ancillary files.

Let $|x_i|$ be the cardinal of (number of elements in) the abstract, countable set, $x_i \in \{x_1, \cdots, x_n\}$. And let the countable value, v_c, be the number of ordered combinations (n-tuples) of $x_1, \cdots, x_n$. The Euclidean volume equation will be proved to be an instance of v_c, where: each domain and range set element corresponds to a value, $\kappa \in \mathbb{R}$, $\kappa = f(|x_i|) : |x_i| \to \infty \Rightarrow \kappa \to 0$ and:

$$v_c = \prod_{i=1}^{n} |x_i| \quad \Rightarrow \quad v = \lim_{\kappa \to 0} v_c \kappa = \lim_{\kappa \to 0} (\prod_{i=1}^{n} |x_i|)\kappa = \cdots = \prod_{i=1}^{n} s_i. \;\; v, s_i \in \mathbb{R}. \quad (1)$$

For all $n > 1$, there are many cases where different domain values, $|x_1|, \cdots, |x_n|$, multiplied yield the same range value, v_c. Inferring a domain value, d_c, from v_c, requires an inverse (bijective) function, $d_c = f_n^{-1}(v_c)$ and $v_c = f_n(d_c)$. The simplest bijective case for $n = 1$ is: $v_c = |x_1| = d_c$. The simplest bijective case for all n that includes the $n = 1$ case is, where $\kappa = f(d_c) : d_c \to \infty \Rightarrow \kappa \to 0$ and:

$$\exists\, d_c, v_c \in \{0, \mathbb{N}\},\; v, d, \kappa \in \mathbb{R} : \;\; v = \lim_{\kappa \to 0} v_c \kappa = \lim_{\kappa \to 0} (\prod_{i=1}^{n} d_c)\kappa = d_c^n \kappa = \cdots = d^n. \quad (2)$$

A set of n-tuples being the union of disjoint subsets of n-tuples implies that the domain value, d_c, is the inverse function of the number of n-tuples, v_c, where v_c is the sum of the

number of subset n-tuples:

$$d_c^n = v_c = \sum_{i=1}^m v_{c_i} = \sum_{i=1}^m (\prod_{j=1}^n |x_{i,j}|) \quad \Rightarrow \quad d^n = \sum_{i=1}^m (\prod_{j=1}^n s_{i,j}). \tag{3}$$

Where each $s_{i,j}$ is $\pm$-signed, the $n = 2$ case is the dot product.

Where each v_{c_i} also is the bijective function, $v_{c_i} = d_{c_i}^n$:

$$d_c^n = \sum_{i=1}^m d_{c_i}^n \quad \Rightarrow \quad d^n = \sum_{i=1}^m d_i^n. \tag{4}$$

$|d|$ is the p-norm (Minkowski distance) [14], which will be proved to imply the metric space properties [10]. The $n = 2$ case is the Euclidean distance.

Many sets contain unordered elements, where there is no intrinsic property making a set element the first, second, $\cdots$, or last (n-th). For example, the set elements, {height, width, depth} $\equiv$ {depth, width, height} are unordered and the set elements, {time, mass, charge} $\equiv$ {mass, charge, time} are unordered.

Sequencing a set of unordered elements reliably in the same order requires that the elements be placed into a "list," which assigns relative sequential positions with labels, for example, labeling the unordered elements as x_1, x_2, $\cdots x_n$. The list can then be sequenced via successor and predecessor functions, $successor(x_i) = x_{i+1}$ and $predecessor(x_i) = x_{i-1}$. For example, the volume successor sequence, $v = \prod_{i=1}^n s_i$, sequences the corresponding set elements, {width, height, depth}, in the same order on each sequencing.

But the union, intersection, multiplication, and addition operations defining the total number of n-tuples and the corresponding volume and distance equations are commutative. The commutative property allows sequencing the *same* totally ordered list of n number of elements in all of the $n!$ possible unique sequences (permutations).

Only a cyclic list, where $successor(x_n) = x_1$ and $predecessor(x_1) = x_n$, allows each element to be the first element of some of those $n!$ permutations. For example, where $n = 3$, the volume cyclic successor sequences are: $v = s_1 \times s_2 \times s_3$. $v = s_2 \times s_3 \times s_1$, $v = s_3 \times s_1 \times s_2$, and the cyclic predecessor sequences are: $v = s_1 \times s_3 \times s_2$. $v = s_2 \times s_1 \times s_3$, $v = s_3 \times s_2 \times s_1$.

The only cyclic lists that allows all $n!$ permutations, using the cyclic successor and predecessor functions, are where each list element is sequentially adjacent (either an *immediate* cyclic successor or an *immediate* cyclic predecessor) to every other element, referred to, here, as an immediate symmetric cyclic list (ISCL). An ISCL will be proved to contain $n \leq 3$ elements.

Application of the ISCL and corresponding lengths to physics:

1. **ISCL:** $\{r_1, r_2, r_3\}$ is an ISCL of 3 "distance" dimensions and $\{t\ (time), m\ (mass), q\ (charge)\}$ is the ISCL of 3 "non-distance" dimensions, each dimension $\subseteq \mathbb{R}$. There is a total of 6 unordered dimensions with the $\subseteq \mathbb{R}$ property: $r_1,\ r_2,\ r_3,\ t,\ m,\ q$.

2. **Corresponding lengths:** Where the dimensions of Euclidean distance, time, mass, and charge are each divided into countably infinite sets of same-sized intervals $\subset \mathbb{R}$ and $i \in \mathbb{N}$, the ith unit interval length, r_p of Euclidean distance r, corresponds 1-1 to the ith unit interval lengths: t_p of time t, m_p mass m, and q_p charge q, where "unit" means a measurement type, for example, meters, seconds, kilograms, and Coulombs.

The 1-1 correspondence of unit interval lengths implies unit ratios of r_p to t_p, m_p, and q_p: $r = (r_p/t_p)t$, $r = (r_p/m_p)m$, and $r = (r_p/q_p)q$, letting: $c_t = r_p/t_p$, $c_m = r_p/m_p$, and $c_q = r_p/q_p$. For example, derivation of the gravity constant, G, will be shown to be $G = c_m c_t^2$, where c_t is the speed of light.

Algebraic manipulation of the 3 direct proportion ratios yields 3 inverse proportion ratios, $rt = t_p r_p$, $rm = m_p r_p$. and $rq = q_p r_p$, letting: $k_t = t_p r_p$, $k_m = m_p r_p$, and $k_q = q_p r_p$. For example, derivation of the reduced Planck constant, $\hbar$, will be shown to be $\hbar = k_m c_t$.

Note that the quantum units: r_p, t_p, m_p, and q_p are derived from:

$$r_p = \sqrt{r_p^2} = \sqrt{c_t k_t} = \sqrt{c_m k_m} = \sqrt{c_q k_q}. \tag{5}$$

And it will be shown that r_p, t_p, m_p, and q_p are the reduced Planck units.

Some published derivations of physics equations are long, with complex math, and many "reasonable" assumptions because the underlying principles are not well understood. As evidence of that assertion, the derivations of well-known gravity, charge, electromagnetic, relativity, quantum physics equations and constants from the unit ratios are shorter, simpler, and use fewer assumptions. All derivations, in this article, are either directly from unit ratios or from equations previously derived from unit ratios.

The inverse proportion unit ratios are used to add quantum extensions Schwarzschild's gravitational time dilation and black hole metric[15][16], Newton's gravity force[2], and Coulomb's charge force[3] equations. The quantum extended equations predict that where quantum effects are measurable, the constants: gravity, G, Einstein's $\kappa = 8\pi G/c^4$[4][5], charge, k_e, permittivity, ε_0, and permeability, μ_0, do not exist. And the quantum-extended equations also predict that as distance between two point masses and between two point charges go to zero, the forces do not go to infinity.

RULER MEASURE AND CONVERGENCE

Definition .1. Ruler measure, $M = \sum_{i=1}^{p} \kappa = p\kappa$, where $\forall\ s, \kappa \in \mathbb{R}, \quad 0 < |\kappa| \leq 1$, $(p = floor(s/\kappa) \quad \vee \quad p = ceiling(s/\kappa))$.

Theorem .2. *Ruler convergence:* $M = \lim_{\kappa \to 0} p\kappa = s$.

The formal proof, "limit_c_0_M_eq_exact_size," is in the file, euclidrelations.v.

Proof. (epsilon-delta proof)
$floor(x)$ is the integer part of x. Therefore:

$$floor(x) = max(\{y :\ y \leq x,\ y \in \mathbb{Z},\ x \in \mathbb{R}\}) \quad \Rightarrow \quad |floor(x) - x| < 1. \tag{6}$$

$$|floor(s/\kappa) - s/\kappa| < 1 \quad \wedge \quad p = floor(s/\kappa) \quad \Rightarrow \quad |p - s/\kappa| < 1. \tag{7}$$

Multiply both sides of inequality 7 by $|\kappa|$:

$$|p - s/\kappa| < 1 \quad \Rightarrow \quad |p\kappa - s| < |\kappa| = |\kappa - 0|. \tag{8}$$

$$\forall\ \epsilon = \delta \quad \wedge \quad |p\kappa - s| < |\kappa - 0| < \delta \quad \Rightarrow \quad |\kappa - 0| < \delta \ \wedge\ |p\kappa - s| < \epsilon$$

$$\stackrel{\text{def}}{=} \quad M = \lim_{\kappa \to 0} p\kappa = s. \quad \square \tag{9}$$

The following is an example of ruler convergence where, $s = \pi \ \Rightarrow\ p = floor(s/\kappa) \Rightarrow p\kappa = 3.1_{\kappa=10^{-1}},\ 3.14_{\kappa=10^{-2}},\ 3.141_{\kappa=10^{-3}}, ...,\ \pi_{\lim_{\kappa \to 0}}$.

Lemma .3. $\forall\ n \geq 1,\ 0 < |\kappa| < 1 : \lim_{\kappa \to 0} \kappa^n = \lim_{\kappa \to 0} \kappa$.

Proof. The formal proof , "lim_c_to_n_eq_lim_c," is in the Rocq file, euclidrelations.v.

$$n \geq 1 \quad \wedge \quad 0 < \kappa \leq 1 \quad \Rightarrow \quad 0 < \kappa^n \leq \kappa \quad \Rightarrow \quad |\kappa - \kappa^n| \leq |\kappa| = |\kappa - 0|. \tag{10}$$

$$\forall\ \epsilon = \delta \quad \wedge \quad |\kappa - \kappa^n| \leq |\kappa - 0| < \delta \quad \Rightarrow \quad |\kappa - 0| < \delta \quad \wedge \quad |\kappa - \kappa^n| < \delta = \epsilon$$

$$\stackrel{\text{def}}{=} \quad \lim_{\kappa \to 0} \kappa^n = 0. \tag{11}$$

$$\lim_{\kappa \to 0} \kappa^n = 0 \quad \wedge \quad \lim_{\kappa \to 0} \kappa = 0 \quad \Rightarrow \quad \lim_{\kappa \to 0} \kappa^n = \lim_{\kappa \to 0} \kappa. \quad \square \tag{12}$$

VOLUME

Euclidean volume

Theorem .4. *Euclidean volume,*

$$\forall\ v_c, d_c, |x_i| \in \{0,\ \mathbb{N}\},\ \ x_i \in \{x_1, \cdots, x_n\},\ v_c = \prod_{i=1}^{n} |x_i| \ \Rightarrow\ v = \prod_{i=1}^{n} s_i, \quad s_i, v \in \mathbb{R}. \tag{13}$$

The formal proof, "Euclidean_volume," is in the Rocq file, euclidrelations.v.

Proof.

$$v_c = \prod_{i=1}^{n} |x_i| \quad \Leftrightarrow \quad v_c\kappa = (\prod_{i=1}^{n} |x_i|)\kappa \quad \Leftrightarrow \quad \lim_{\kappa\to 0} v_c\kappa = \lim_{\kappa\to 0}(\prod_{i=1}^{n} |x_i|)\kappa. \tag{14}$$

Apply the ruler (.1) and ruler convergence (.2) to equation 14:

$$\exists\, v, \kappa \in \mathbb{R} :\ v_c = floor(v/\kappa) \quad \Rightarrow \quad v = \lim_{\kappa\to 0} v_c\kappa \tag{15}$$

$$v = \lim_{\kappa\to 0} v_c\kappa \quad \wedge \quad \lim_{\kappa\to 0} v_c\kappa = \lim_{\kappa\to 0}(\prod_{i=1}^{n} |x_i|)\kappa \Rightarrow \quad v = \lim_{\kappa\to 0}(\prod_{i=1}^{n} |x_i|)\kappa. \tag{16}$$

Apply lemma .3 to equation 15:

$$v = \lim_{\kappa\to 0}(\prod_{i=1}^{n} |x_i|)\kappa \quad \wedge \quad \lim_{\kappa\to 0} \kappa^n = \lim_{\kappa\to 0} \kappa$$
$$\Rightarrow \quad v = \lim_{\kappa\to 0}(\prod_{i=1}^{n} |x_i|)\kappa^n = \lim_{\kappa\to 0}(\prod_{i=1}^{n}(|x_i|\kappa)). \tag{17}$$

Apply the ruler (.1) and ruler convergence (.2) to s_i:

$$\exists\, s_i, \kappa \in \mathbb{R} :\ floor(s_i/\kappa) = |x_i| \quad \Rightarrow \quad \lim_{\kappa\to 0}(|x_i|\kappa) = s_i. \tag{18}$$

$$v = \lim_{\kappa\to 0}(\prod_{i=1}^{n} |x_i|\kappa) \quad \wedge \quad \lim_{\kappa\to 0}(|x_i|\kappa) = s_i \quad \Leftrightarrow \quad v = \prod_{i=1}^{n} s_i \quad \square \tag{19}$$

Lemma .5. *The number of n-tuples, v_c, is the sum of the number of n-tuples, v_{c_i}, in each subset of n-tuples, implies a volume is the sum of volumes,*

$$v_c = \sum_{i=1}^{m} v_{c_i} \quad \Rightarrow \quad v = \sum_{i=1}^{m} v_i, \quad v, v_i \in \mathbb{R}.$$

The formal proof, "sum_of _volumes," is in the Rocq file, euclidrelations.v.

Proof. From the condition of this theorem:

$$v_c = \sum_{i=1}^{m} v_{c_i} \quad \Leftrightarrow \quad \lim_{\kappa\to 0} v_c\kappa = \lim_{\kappa\to 0} \sum_{j=1}^{m}(v_{c_i}\kappa). \tag{20}$$

Apply lemma .3 to equation 20:

$$\lim_{\kappa\to 0} v_c\kappa = \lim_{\kappa\to 0}(\sum_{j=1}^{m} v_{c_i})\kappa \quad \wedge \quad \lim_{\kappa\to 0} \kappa^n = \lim_{\kappa\to 0} \kappa$$
$$\Leftrightarrow \quad \lim_{\kappa\to 0} v_c\kappa = \lim_{\kappa\to 0} \sum_{j=1}^{m}(v_{c_i}\kappa). \tag{21}$$

Apply the ruler (.1) and ruler convergence theorem (.2) to equation 21:

$$\exists\, v, v_i :\ v = \lim_{\kappa\to 0} v_c\kappa \quad \wedge \quad \lim_{\kappa\to 0} v_c\kappa = \lim_{\kappa\to 0} \sum_{j=1}^{m}(v_{c_i}\kappa)$$
$$\Rightarrow \quad v = \lim_{\kappa\to 0} \sum_{j=1}^{m}(v_{c_i}\kappa). \tag{22}$$

Apply the ruler (.1) and ruler convergence theorem (.2) to equation 22:

$$v = \lim_{\kappa\to 0} \sum_{j=1}^{m} (v_{c_i}\kappa) \quad \wedge \quad \exists\, v_i, v_{c_i} :\ v_i = \lim_{\kappa\to 0} v_{c_i}\kappa \quad \Rightarrow \quad v = \sum_{j=1}^{m} v_i. \tag{23}$$

□

DISTANCE

Definition .6. Bijective, countable domain value, d_c:

$$\exists\, v_c, d_c, |x_i| \in \{0,\ \mathbb{N}\},\ \ x_i \in \{x_1, \cdots, x_n\}, \quad v_c = \prod_{i=1}^{n} |x_i| = \prod_{i=1}^{n} d_c = d_c^n. \tag{24}$$

Vector inner product

Theorem .7. *Sum of volumes distance:*

$$d_c^n = v_c = \sum_{i=1}^{m} v_{c_i} \quad \Rightarrow \quad d^n = v = \sum_{i=1}^{m} (\prod_{j=1}^{n} s_{i_j}).$$

The formal proof, "sum_of _volumes _distance," is in the Rocq file, euclidrelations.v.

Proof. Multiply both sides of the assumption by κ and take the limit:

$$d_c^n = v_c \quad \Rightarrow \quad \lim_{\kappa\to 0} d_c^n\kappa = \lim_{\kappa\to 0} v_c\kappa. \tag{25}$$

Apply lemma (.3):

$$\lim_{\kappa\to 0} d_c^n\kappa = \lim_{\kappa\to 0} v_c\kappa \ \wedge\ \lim_{\kappa\to 0} \kappa^n = \lim_{\kappa\to 0} \kappa \ \Rightarrow\ \lim_{\kappa\to 0}(d_c\kappa)^n = \lim_{\kappa\to 0} v_c\kappa. \tag{26}$$

Apply the ruler (.1) and ruler convergence (.2) to d and v:

$$\exists\, d, v, \kappa \in \mathbb{R} :\ floor(d/\kappa) = d_c \ \wedge\ floor(v/\kappa) = v_c \ \wedge\ \lim_{\kappa\to 0}(d_c\kappa)^n = \lim_{\kappa\to 0} v_c\kappa$$
$$\Rightarrow \quad d^n = \lim_{\kappa\to 0}(d_c\kappa)^n = \lim_{\kappa\to 0} v_c\kappa = v. \tag{27}$$

From the sum of volumes lemma .5 and the Euclidean volume theorem .4:

$$d^n = v = \sum_{i=1}^{m} v_i \quad \wedge \quad v_i = \prod_{j=1}^{n} s_{i_j} \quad \Rightarrow \quad d^n = v = \sum_{i=1}^{m} v_i = \sum_{i=1}^{m} (\prod_{j=1}^{n} s_{i_j}). \tag{28}$$

□

Note: In the volume proof (.4), where κ is negative, volume is negative. From the lemma (.3), $\forall\ n \geq 1,\ 0 < |\kappa| < 1 : \lim_{\kappa\to 0} \kappa^n = \lim_{\kappa\to 0} \kappa,\ \exists\ \kappa_i \in \{\kappa_1, \cdots, \kappa_n\}$. Therefore, κ_i can have negative values, where $\prod_{i=1}^{n} \kappa_i = \kappa^n$. And where $\kappa_i < 0$, the corresponding $s_{i,j} < 0$. Therefore, the $n = 2$ case of the sum of volumes distance is the dot product. The Euclidean space dot product is generalized to the inner product by allowing each $s_{i,j}$ term to be a function or complex number.

Minkowski distance (p-norm)

Theorem .8. *Minkowski distance (p-norm):*

$$d_c^n = v_c = \sum_{i=1}^m v_{c_i} = \sum_{i=1}^m d_{c_i}^n \quad \Leftrightarrow \quad d^n = \sum_{i=1}^m d_i^n.$$

The formal proof, "Minkowski_distance," is in the Rocq file, euclidrelations.v.

Proof. From the sum of volumes distance theorem .7:

$$d_c^n = v_c = \sum_{i=1}^m v_{c_i} \quad \Rightarrow \quad d^n = v = \sum_{i=1}^m v_i \tag{29}$$

$$d^n = v = \sum_{i=1}^m v_i \quad \wedge \quad d_i^n = v_i \Rightarrow \quad d^n = \sum_{i=1}^m d_i^n \quad \square \tag{30}$$

Theorem .9. *Distance triangle inequality*

$$\forall\, n \in \mathbb{N},\ \ v_a, v_b \geq 0 :\ \ (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n}.$$

The formal proof, distance_inequality, is in the Rocq file, euclidrelations.v.

Proof. Expand $(v_a^{1/n} + v_b^{1/n})^n$ using the binomial expansion:

$$\forall\, v_a, v_b \geq 0 : \quad v_a + v_b \leq v_a + v_b + \sum_{i=1}^n \binom{n}{k} (v_a^{1/n})^{n-k} (v_b^{1/n})^k + \sum_{i=1}^n \binom{n}{k} (v_a^{1/n})^k (v_b^{1/n})^{n-k} = (v_a^{1/n} + v_b^{1/n})^n. \tag{31}$$

Take the nth root of both sides of the inequality 31:

$$\forall\, v_a, v_b \geq 0,\ \ n \in \mathbb{N} : \quad v_a + v_b \leq (v_a^{1/n} + v_b^{1/n})^n \quad \Rightarrow \quad (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n}. \tag{32}$$

$\square$

Theorem .10. *Distance sum inequality*

$$\forall\, m, n \in \mathbb{N},\ \ a_i, b_i \geq 0 : (\sum_{i=1}^m (a_i^n + b_i^n))^{1/n} \leq (\sum_{i=1}^m a_i^n)^{1/n} + (\sum_{i=1}^m b_i^n)^{1/n}.$$

The formal proof, distance_sum_inequality, is in the Rocq file, euclidrelations.v.

Proof. Apply the distance triangle inequality (.9):

$$\forall\, m, n \in \mathbb{N},\ \ v_a, v_b \geq 0 :\ \ v_a = \sum_{i=1}^m a_i^n \quad \wedge \quad v_b = \sum_{i=1}^m b_i^n \quad \wedge \quad (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n}$$
$$\Rightarrow \quad ((\sum_{i=1}^m a_i^n) + (\sum_{i=1}^m b_i^n))^{1/n} = (\sum_{i=1}^m (a_i^n + b_i^n))^{1/n} \leq (\sum_{i=1}^m a_i^n)^{1/n} + (\sum_{i=1}^m b_i^n)^{1/n}. \quad \square \tag{33}$$

Metric Space

All Minkowski distances (p-norms) imply the metric space properties. The formal proofs: triangle_inequality, symmetry, non_negativity, and identity_of_indiscernibles are in the Rocq file, euclidrelations.v.

Theorem .11. *Triangle Inequality:*

$$d(s_1, s_2) = (\sum_{i=1}^{2} s_i^p)^{1/p}, \quad p \geq 1, \quad u = s_1, \quad w = s_2, \quad v = w/k$$
$$\Rightarrow \quad d(u, w) = (u^p + w^p)^{1/p} \leq (u^p + v^p)^{1/p} + (v^p + w^p)^{1/p} = d(u, v) + d(v, w).$$

Proof. $\forall\, p \geq 1, \quad u = s_1, \quad w = s_2, \quad v = w/k$:

$$(u^p + w^p)^{1/p} \leq ((u^p + w^p) + 2v^p)^{1/p} = ((u^p + v^p) + (v^p + w^p))^{1/p}. \tag{34}$$

Apply the distance triangle inequality (.9) to the inequality 34:

$$(u^p + w^p)^{1/p} \leq ((u^p + v^p) + (v^p + w^p))^{1/p} \quad \wedge \quad (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n}$$
$$\wedge \quad v_a = u^p + v^p \quad \wedge \quad v_b = v^p + w^p$$
$$\Rightarrow \quad (u^p + w^p)^{1/p} \leq ((u^p + v^p) + (v^p + w^p))^{1/p} \leq (u^p + v^p)^{1/p} + (v^p + w^p)^{1/p}$$
$$\Rightarrow \quad d(u, w) = (u^p + w^p)^{1/p} \leq (u^p + v^p)^{1/p} + (v^p + w^p)^{1/p} = d(u, v) + d(v, w). \quad \square \tag{35}$$

Theorem .12. *Symmetry:*
$d(s_1, s_2) = (\sum_{i=1}^{2} s_i^p)^{1/p}, \quad p \geq 1, \quad u = s_1, \quad v = s_2 \quad \Rightarrow \quad d(u, v) = d(v, u).$

Proof. By the commutative law of addition:

$$\forall\, p \,:\, p \geq 1, \quad d(s_1, s_2) = (\sum_{i=1}^{2} s_i^p)^{1/p} = (s_1^p + s_2^p)^{1/p}$$
$$\Rightarrow \quad d(u, v) = (u^p + v^p)^{1/p} = (v^p + u^p)^{1/p} = d(v, u). \quad \square \tag{36}$$

Theorem .13. *Non-negativity:*

$$d(s_1, s_2) = (\sum_{i=1}^{2} s_i^p)^{1/p}, \quad p \geq 1, \quad u = s_1, \quad w = s_2, \quad v = w/k \quad \Rightarrow \quad d(u, w) \geq 0.$$

Proof. By definition, the length of an interval is always ≥ 0:

$$\forall\, [a_1, b_1],\ [a_2, b_2], \quad u = b_1 - a_1,\ v = b_2 - a_2, \quad \Rightarrow \quad u \geq 0,\ v \geq 0. \tag{37}$$
$$p \geq 1,\ \ u, v \geq 0 \quad \Rightarrow \quad d(u, v) = (u^p + v^p)^{1/p} \geq 0. \quad \square \tag{38}$$

Theorem .14. *Identity of Indiscernibles:* $d(u, u) = 0$.

Proof. From the non-negativity property (.13):

$$d(u, w) \geq 0 \quad \wedge \quad d(u, v) \geq 0 \quad \wedge \quad d(v, w) \geq 0 \tag{39}$$

$$\Rightarrow \quad \exists\, d(u, w) = d(u, v) = d(v, w) = 0. \tag{40}$$

$$d(u, w) = d(v, w) = 0 \quad \Rightarrow \quad u = v. \tag{41}$$

$$d(u, v) = 0 \quad \wedge \quad u = v \quad \Rightarrow \quad d(u, u) = 0. \quad \square \tag{42}$$

Property limiting a set to at most 3 elements

The following definitions and proof use first order logic. A Horn clause-like expression is used, here, to make the proof easier to read. By convention, the proof goal, in a Horn clause, is on the left side of the implication sign ($\leftarrow$) and supporting facts are on the right side. The formal proofs in the Rocq file threed.v are: adj111, adj122, adj212, adj123, adj133, adj233, adj213, adj313, adj323, and not_all_mutually_adjacent_gt_3.

Definition .15. Immediate Cyclic Successor of m is n:

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\} :$$
$$Successor(m, n, setsize) \quad \leftarrow \quad (m = setsize \wedge n = 1) \quad \vee \quad (n = m + 1 \leq setsize). \tag{43}$$

Definition .16. Immediate Cyclic Predecessor of m is n:

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\} :$$
$$Predecessor(m, n, setsize) \quad \leftarrow \quad (m = 1 \wedge n = setsize) \quad \vee \quad (n = m - 1 \geq 1). \tag{44}$$

Definition .17. Adjacent: element m is sequentially adjacent to element n if the immediate cyclic successor of m is n or the immediate cyclic predecessor of m is n. Notionally:

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\} :$$
$$Adjacent(m, n, setsize) \leftarrow Successor(m, n, setsize) \;\vee\; Predecessor(m, n, setsize). \tag{45}$$

Definition .18. Immediate Symmetric Cylic List (ISCL), where every set element is sequentially adjacent to every other element):

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\} : \quad Adjacent(m, n, setsize). \tag{46}$$

Theorem .19. *An ISCL is limited to at most 3 elements.*

$$\forall\, x_m, x_n \in \{x_1, \cdots, x_{setsize}\}: \quad Adjacent(m, n, setsize) \quad \Rightarrow \quad setsize \leq 3. \tag{47}$$

Proof.

Every element is adjacent to every other element, where $setsize \in \{1,\ 2,\ 3\}$:

$$Adjacent(1,1,1) \leftarrow Successor(1,1,1) \leftarrow (m = setsize \wedge n = 1). \tag{48}$$

$$Adjacent(1,2,2) \leftarrow Successor(1,2,2) \leftarrow (n = m + 1 \leq setsize). \tag{49}$$

$$Adjacent(1,2,3) \leftarrow Successor(1,2,3) \leftarrow (n = m + 1 \leq setsize). \tag{50}$$

$$Adjacent(2,1,3) \leftarrow Predecessor(2,1,3) \leftarrow (n = m - 1 \geq 1). \tag{51}$$

$$Adjacent(3,1,3) \leftarrow Successor(3,1,3) \leftarrow (n = setsize \wedge m = 1). \tag{52}$$

$$Adjacent(1,3,3) \leftarrow Predecessor(1,3,3) \leftarrow (m = 1 \wedge n = setsize). \tag{53}$$

$$Adjacent(2,3,3) \leftarrow Successor(2,3,3) \leftarrow (n = m + 1 \leq setsize). \tag{54}$$

$$Adjacent(3,2,3) \leftarrow Predecessor(3,2,3) \leftarrow (n = m - 1 \geq 1). \tag{55}$$

Element 2 is the only immediate successor of element 1 for all $setsize \geq 3$, which implies element 3 is not ($\neg$) an immediate successor of element 1 for all $setsize \geq 3$:

$$\neg Successor(1,3,setsize \geq 3) \leftarrow Successor(1,2,setsize \geq 3) \leftarrow (n = m + 1 \leq setsize). \tag{56}$$

Element $n = setsize > 3$ is the only immediate predecessor of element 1, which implies element 3 is not ($\neg$) an immediate predecessor of element 1 for all $setsize > 3$:

$$\neg Predecessor(1,3,setsize \geq 3) \leftarrow Predecessor(1,setsize,setsize > 3) \leftarrow (m = 1 \wedge n = setsize > 3). \tag{57}$$

For all $setsize > 3$, some elements are not ($\neg$) sequentially adjacent to every other element (not immediate symmetric):

$$\neg Adjacent(1,3,setsize > 3) \leftarrow \neg Successor(1,3,setsize > 3) \quad \wedge \quad \neg Predecessor(1,3,setsize > 3). \quad \square \tag{58}$$

The Symmetric goal matches the Adjacent goals 48 and fails for all "setsize" greater than three.

DERIVATION OF PHYSICS EQUATIONS AND CONSTANTS

The unit ratios

Application of the ISCL (.19) and properties of Euclidean geometry (.4) to physics:

1. **ISCL:** $\{r_1, r_2, r_3\}$ is an ISCL of 3 "distance" dimensions and $\{t\ (time), m\ (mass), q\ (charge)\}$ is the ISCL of 3 "non-distance" dimensions, each dimension $\subseteq \mathbb{R}$. There is a total of 6 unordered dimensions with the $\subseteq \mathbb{R}$ property: $r_1,\ r_2,\ r_3,\ t,\ m,\ q$.

2. **Corresponding lengths:** Where the dimensions of Euclidean distance, time, mass, and charge are each divided into countably infinite sets of same-sized intervals $\subset \mathbb{R}$ and $i \in \mathbb{N}$, the ith unit interval length, r_p of Euclidean distance r, corresponds 1-1 to the ith unit interval lengths: t_p of time t, m_p mass m, and q_p charge q, where "unit" means a measurement type, for example, meters, seconds, kilograms, and Coulombs.

The 1-1 correspondence of unit interval lengths implies unit ratios of r_p to t_p, m_p, and q_p. A corresponding number of units can be expressed as, $r/r_p = \tau/\tau_p$, where $\tau \in \{t, m, q\}$. Multiplying both sides by r_p yields 3 direct proportion ratios:

$$\forall\, r_p, \tau_p, r, \tau \subseteq \mathbb{R},\ r_p, \tau_p\ \textbf{constants},\ \tau \in \{t, m, q\} :\ r/r_p = \tau/\tau_p \quad \Rightarrow \quad r = (r_p/\tau_p)/\tau$$
$$\Rightarrow \quad r = (r_p/t_p)t \quad \wedge \quad r = (r_p/m_p)m \quad \wedge \quad r = (r_p/q_p)q. \tag{59}$$

And for conciseness, let:

$$r_p/t_p = c_t \quad \wedge \quad r_p/m_p = c_m \quad \wedge \quad r_p/q_p = c_q. \tag{60}$$

Algebraic manipulation of the 3 direct proportion ratios yields 3 inverse proportion ratios:

$$r/t = r_p/t_p \quad \wedge \quad r/m = r_p/m_p \quad \Rightarrow \quad (r/t)/(r/m) = (r_p/t_p)/(r_p/m_p)$$
$$\Rightarrow \quad (mr)/(tr) = (m_p r_p)/(t_p r_p) \quad \Rightarrow \quad \exists\, r, t, m \in \mathbb{R} :\ mr = m_p r_p, \quad tr = t_p r_p. \tag{61}$$

$$r/t = r_p/t_p \quad \wedge \quad r/q = r_p/q_p \quad \Rightarrow \quad (r/t)/(r/q) = (r_p/t_p)/(r_p/q_p)$$
$$\Rightarrow \quad (qr)/(tr) = (q_p r_p)/(t_p r_p) \quad \Rightarrow \quad \exists\, r, t, q \in \mathbb{R} :\ qr = q_p r_p, \quad tr = t_p r_p. \tag{62}$$

And for conciseness, let:

$$t_p r_p = k_t \quad \wedge \quad m_p r_p = k_m \quad \wedge \quad q_p r_p = k_q. \tag{63}$$

G, Newton, Gauss, Poisson gravity laws

From equations 60, linear acceleration, r/t^2, can be related to mass, m, and distance, r:

$$r = c_m m \;\wedge\; r = c_t t \;\Rightarrow\; r/(c_t t)^2 = c_m m/r^2 \;\Rightarrow\; r/t^2 = (c_m c_t^2)m/r^2 = Gm/r^2, \tag{64}$$

where $G = c_m c_t^2$, conforms to the SI units: $m^3 \cdot kg^{-1} \cdot s^{-2}$ [2].

Newton's law follows from multiplying both sides of equation 64 by m:

$$r/t^2 = Gm/r^2 \;\Leftrightarrow\; F \stackrel{\text{def}}{=} mr/t^2 = Gm^2/r^2. \tag{65}$$

$$F = Gm^2/r^2 \;\wedge\; \forall\, m \in \mathbb{R} :\; \exists\, m_1, m_2 \in \mathbb{R} :\; m_1 m_2 = m^2 \quad\Rightarrow\quad F = Gm_1 m_2/r^2. \tag{66}$$

A new interpretation of Gauss's and Poisson's gravity laws is presented here. Equation 64 relates linear acceleration, r/t^2, to mass and distance. Gauss's gravity field, $\mathbf{g}$, and Poisson's gravity field, $-\nabla\Phi(\tilde{\mathbf{r}}, t)$ is the angular (orbital) acceleration, $2\pi r/t^2$. Applying equation 64:

$$r/t^2 = Gm/|\tilde{\mathbf{r}}|^2 \;\wedge\; \mathbf{g} = -\nabla\Phi(\tilde{\mathbf{r}}, t) = 2\pi r/t^2 \quad\Rightarrow\quad \mathbf{g} = -\nabla\Phi(\tilde{\mathbf{r}}, t) = 2\pi Gm/|\tilde{\mathbf{r}}|^2 \tag{67}$$

$$\mathbf{g} = -\nabla\Phi(\tilde{\mathbf{r}}, t) = -2\pi Gm/|\tilde{\mathbf{r}}|^2 \quad\Rightarrow\quad \nabla\cdot\mathbf{g} = \nabla^2\Phi(\tilde{\mathbf{r}}, t) = -4\pi Gm/|\tilde{\mathbf{r}}|^3 = -4\pi G\rho, \tag{68}$$

where $\rho = m/|\tilde{\mathbf{r}}|^3$ is the mass density:

Special relativity

$$\forall\, \tau \in \{t, m, q\},\; r^2 = r'^2 + r_v^2,\; \exists\, \mu, \nu :\; r = \mu\tau \quad\wedge\quad r_v = \nu\tau \quad\Rightarrow\quad (\mu\tau)^2 = r'^2 + (\nu\tau)^2$$
$$\Rightarrow\quad r' = \sqrt{(\mu\tau)^2 - (\nu\tau)^2} = \mu\tau\sqrt{1 - (\nu/\mu)^2}. \tag{69}$$

Local frame distance, r', contracts relative to a distant observer frame distance, r, as $\nu \to \mu$:

$$r' = \mu\tau\sqrt{1 - (\nu/\mu)^2} \quad\wedge\quad \mu\tau = r \quad\Rightarrow\quad r' = r\sqrt{1 - (\nu/\mu)^2}. \tag{70}$$

A distant observer frame type, τ, dilates relative to the local observer frame type, τ', as $\nu \to \mu$:

$$\mu\tau = r'/\sqrt{1 - (\nu/\mu)^2} \quad\wedge\quad r' = \mu\tau \quad\Rightarrow\quad \tau = \tau'/\sqrt{1 - (\nu/\mu)^2}. \tag{71}$$

Where τ is type, time, the space-like flat Minkowski spacetime event interval is:

$$\mathrm{d}r^2 = \mathrm{d}r'^2 + \mathrm{d}r_v^2 \quad\wedge\quad \mathrm{d}r_v^2 = \mathrm{d}r_1^2 + \mathrm{d}r_2^2 + \mathrm{d}r_3^2 \quad\wedge\quad \mathrm{d}(\mu\tau) = \mathrm{d}r$$
$$\Rightarrow\; \mathrm{d}r'^2 = \mathrm{d}(\mu\tau)^2 - \mathrm{d}r_1^2 - \mathrm{d}r_2^2 - \mathrm{d}r_3^2. \tag{72}$$

Simple general relativity solutions

The Einstein field equation (EFE) is:

$$G_{\mu,\nu} + \Lambda g_{\mu,\nu} = \mathbf{R} + \frac{1}{2} R g_{\mu,\nu} + \Lambda g_{\mu,\nu} = \kappa T_{\mu,\nu}, \tag{73}$$

where: $G_{\mu,\nu}$ is Einstein's tensor, $\mathbf{R}$ is the Ricci curvatature, R is the scalar curvature, $g_{\mu,\nu}$ is the metric tensor, Λ is a cosmological constant, $\kappa = 8\pi G/c^4$ is Einstein's constant, and $T_{\mu,\nu}$ is the stress-energy-momentum tensor [6][5].

The EFE is nonlinear, which has made determining the measure (metric) of the spacetime curvature, $g_{\mu,\nu}$, from the distribution of the stress-energy-momentum, $T_{\mu,\nu}$ very difficult. However, determining $g_{\mu,\nu}$ is very simple by noting that: 1) The infinitesimal space near every local coordinate point in the EFE is Euclidean-like[6][5]; 2) The unit ratios are with respect to Euclidean distance and, therefore, all physics equations (laws) derived from the unit ratios are the same at every Euclidean-like local coordinate point; and 3) The spacetime curvature is specified by one function for each diagonal component of the metric, $g_{\mu,\nu}$ [6].

This leads to the following simple steps to solve for the metric tensor, $g_{\mu,\nu}$, independent of the EFE. The next subsection contains an example of using these steps for a much shorter and simpler derivation of Schwarzschild's black hole metric [15][16].

Step 1) Use the unit ratios and other equations derived from the ratios, for example, the special relativity equations (70) and the equation defining the constant, G (64), to derive functions to be assigned to the diagonal components of $g_{\nu,\mu}$.

Step 2) The 4D flat spacetime interval equation is an instance of the 2D equation (72), $\mathrm{d}r'^2 = \mathrm{d}(ct)^2 - \mathrm{d}r_v^2$. Derive the metric first as a 2D metric and expand later to a 4D metric.

(Optional step 2d) The 2D metric tensor allows using the much simpler 2D Ricci curvature and scalar curvature. And the 2D tensors reduce the number of independent equations to solve. The 2D solutions can next be used to set constraints on the solutions in the 4D tensors.

Step 3) One simple method to translate from 2D to 4D is to use spherical coordinates, where the first 2 diagonal components of $g_{\nu,\mu}$ remain unchanged and the other 2 diagonal components are functions of the angles, longitude, θ, and latitude, ϕ.

Schwarzschild's time dilation and metric

From equations 70 and 59:

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-(1)(v^2/c^2)} \quad \wedge \quad c_m m/r = 1$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-c_m m v^2/rc^2}. \quad (74)$$

Where v_{escape} is the escape velocity:

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-c_m m v^2/rc^2} \quad \wedge \quad KE = mv^2/2 = mv_{escape}^2$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m m v_{escape}^2/rc^2}. \quad (75)$$

For a photon, the escape velocity, $v_{escape} = c$.

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m m v_{escape}^2/rc^2}. \quad \wedge \quad v_{escape} = c$$
$$\Rightarrow \quad \sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m mc^2/rc^2}. \quad (76)$$

Combining equation 76 with the derivation of G (64):

$$c_m c^2 = G \wedge \sqrt{1-(v^2/c^2)} = \sqrt{1-2c_m mc^2/rc^2} \Rightarrow \sqrt{1-(v^2/c^2)} = \sqrt{1-2Gm/rc^2}. \quad (77)$$

Combining equation 77 with equation 71 yields Schwarzschild's gravitational time dilation:

$$\sqrt{1-(v^2/c^2)} = \sqrt{1-2Gm/rc^2} \wedge t' = t\sqrt{1-(v^2/c^2)} \Rightarrow t' = t\sqrt{1-2Gm/rc^2}. \quad (78)$$

From equations 70 and 77, where Schwarzschild defined the black hole event horizon radius as $\alpha \stackrel{\text{def}}{=} 2Gm/c^2$ [15] [16]:

$$r' = r\sqrt{1-(v/c)^2} = r\sqrt{1-2Gm/rc^2} \quad \wedge \quad \alpha \stackrel{\text{def}}{=} 2Gm/c^2 \quad \Rightarrow \quad r' = r\sqrt{1-\alpha/r}. \quad (79)$$

Applying equation 79 to the time-like spacetime interval equation 72, where $\mathrm{d}s^2 < 0$:

$$r' = r\sqrt{1-\alpha/r} \quad \wedge \quad \mathrm{d}s^2 = \mathrm{d}r'^2 - \mathrm{d}r^2$$
$$\Rightarrow \quad \mathrm{d}s^2 = (\sqrt{1-\alpha/r}\mathrm{d}r)^2 - (\mathrm{d}r'/\sqrt{1-\alpha/r})^2 = (1-\alpha/r)\mathrm{d}r^2 - (1-\alpha/r)^{-1}\mathrm{d}r'^2. \quad (80)$$
$$\mathrm{d}s^2 = (1-\alpha/r)\mathrm{d}r^2 - (1-\alpha/r)^{-1}\mathrm{d}r'^2 \quad \wedge \quad \mathrm{d}r = \mathrm{d}(ct) \quad \wedge \quad c = 1 \quad \wedge$$
$$\lim_{\mathrm{d}s\to 0} \mathrm{d}r' = \lim_{\mathrm{d}s\to 0} \mathrm{d}r \quad \Rightarrow \quad \lim_{\mathrm{d}s\to 0} \mathrm{d}s^2 = (1-\alpha/r)\mathrm{d}t^2 - (1-\alpha/r)^{-1}\mathrm{d}r^2. \quad (81)$$

Using spherical coordinates to translate from 2D to 4D yields the $+ - --$ form of Schwarzschild's black hole metric [15] [16]:

$$\begin{aligned} \mathrm{d}s^2 &= (1-\alpha/r)\mathrm{d}t^2 - (1-\alpha/r)^{-1}\mathrm{d}r^2 \\ &\Rightarrow \quad \mathrm{d}s^2 = (1-\alpha/r)\mathrm{d}t^2 - (1-\alpha/r)^{-1}\mathrm{d}r^2 - r^2(\mathrm{d}\theta^2 + sin^2\theta\mathrm{d}\phi^2) \\ &\Rightarrow \quad g_{\mu,\nu} = diag[(1-\alpha/r),\ -(1-\alpha/r)^{-1},\ -r^2(\mathrm{d}\theta^2),\ -r^2(sin^2\theta\mathrm{d}\phi^2)]. \end{aligned} \tag{82}$$

k_e, ε_0, E, Coulomb and Gauss laws

From equations 60, linear acceleration, r/t^2, can be related to charge, q, and distance, r:

$$r = c_q q \ \wedge\ r = c_t t \ \Rightarrow\ r/(c_t t)^2 = c_q q/r^2 \ \Rightarrow\ r/t^2 = (c_q c_t^2)q/r^2. \tag{83}$$

$$r = c_m m = c_q q \quad \Rightarrow \quad m = (c_q/c_m)q. \tag{84}$$

Combining equations 83 and 84 yields Coulombs's law:

$$r/t^2 = (c_q c_t^2)q/r^2 \ \wedge\ m = (c_q/c_m)q \ \Rightarrow\ F \stackrel{\text{def}}{=} mr/t^2 = (c_q^2 c_t^2/c_m)q^2/r^2 = k_e q^2/r^2. \tag{85}$$

where $k_e = c_q^2 c_t^2/c_m$, conforms to the SI units: $kg \cdot m^3 \cdot s^{-2} \cdot C^{-2} = N \cdot m^2 \cdot C^{-2}$ [3].

$$\forall\, q \in \mathbb{R}\ \exists\, q_1, q_2 \in \mathbb{R}:\ q_1 q_2 = q^2 \quad \wedge \quad F = k_e q^2/r^2 \quad \Rightarrow \quad F = k_e q_1 q_2/r^2. \tag{86}$$

A new interpretation of Gauss's electric field law is presented here. Equation 83 relates the linear acceleration, r/t^2, to charge and distance. Gauss's electric field, **E**, relates angular (orbital) acceleration, $2\pi r/t^2$ to charge and distance. Applying equation 85:

$$F_C = mr/t^2 = k_e q^2/r^2 \quad \Rightarrow \quad \exists\, F_E \in \mathbb{R}:\ F_E = 2\pi(mr/t^2) = 2\pi k_e q^2/r^2. \tag{87}$$

$$F_E = 2\pi k_e q^2/r^2 = q(2\pi(k_e q/r^2)) \quad \wedge \quad \exists\, E \in R:\ E = 2\pi k_e q/r^2 \quad \Rightarrow \quad F_E = qE. \tag{88}$$

The electric field, $E = 2\pi k_e q/r^2$, conforms to the SI units $kg \cdot m \cdot s^{-2} \cdot C^{-1} = N \cdot C^{-1}$.

$$\mathbf{E} = 2\pi k_e q/|\vec{\mathbf{r}}|^2 \quad \Rightarrow \quad \nabla \cdot \mathbf{E} = -4\pi k_e q/|\vec{\mathbf{r}}|^3. \tag{89}$$

$$\nabla \cdot \mathbf{E} = -4\pi k_e q/|\vec{\mathbf{r}}|^3 \quad \wedge \quad \varepsilon_0 \stackrel{\text{def}}{=} 1/4\pi k_e \quad \wedge \quad \rho \stackrel{\text{def}}{=} q/|\vec{\mathbf{r}}|^3 \quad \Rightarrow \quad \nabla \cdot \mathbf{E} = -\rho/\varepsilon_0, \tag{90}$$

which is Gauss's electric field law [3] in differential form.

B, μ_0, and Lorentz's law

Applying the distance contraction equation, 70, to equation 88, where r' is the distant observer frame of reference and r is local (rest) frame of reference:

$$r' = r/\sqrt{1 - v^2/c^2} \quad \wedge \quad F' = 2\pi k_e q^2/r'^2 \quad \Rightarrow \quad F = 2\pi k_e q^2(1 - v^2/c^2)/r^2. \tag{91}$$

From equation 88:

$$E = 2\pi k_e q/r^2 \ \wedge \ F = 2\pi k_e q^2(1 - v^2/c^2)/r^2 \ \Rightarrow \ F = q(E - (((2\pi k_e/c^2)q/r^2))v^2). \tag{92}$$

$$F = q(E - (((2\pi k_e/c^2)q/r^2))v^2) \ \wedge \ \exists\, B: \ B = (2\pi k_e/c^2)vq/r^2 \ \Rightarrow \ F = q(E - Bv). \tag{93}$$

Translating equation 93 to vector form:

$$F = q(E - Bv) \quad \Rightarrow \quad \mathbf{F} = q(\mathbf{E} - \mathbf{B} \times \vec{\mathbf{v}}). \tag{94}$$

$$\mathbf{B} \times \vec{\mathbf{v}} = -(\vec{\mathbf{v}} \times \mathbf{B})) \quad \wedge \quad \mathbf{F} = q(\mathbf{E} - \mathbf{B} \times \vec{\mathbf{v}}) \quad \Rightarrow \quad \mathbf{F} = q(\mathbf{E} + \vec{\mathbf{v}} \times \mathbf{B}), \tag{95}$$

which is Lorentz law in the distant observer frame of reference, where the magnetic field, $B = (2\pi k_e/c^2)vq/r^2$, conforms to the base SI units: $kg \cdot s^{-1} \cdot C^{-1} \stackrel{\text{def}}{=} kg \cdot s^{-2} \cdot A^{-1} \stackrel{\text{def}}{=} T$.

Based on the assumption (having no math proof-motivation) of a magnetic dipole, the magnetic field divergence outward, $\nabla \cdot \mathbf{B}$, at a point, P, normal to a surface enclosing a charge equals the field divergence inward, $\nabla \cdot \mathbf{B}$:

$$\mathbf{B} = (2\pi k_e/c^2)vq/|\vec{\mathbf{r}}|^2 \quad \Rightarrow \quad \nabla \cdot \mathbf{B} = -(4\pi k_e/c^2)q/|\vec{\mathbf{r}}|^3 + (4\pi k_e/c^2)q/|\vec{\mathbf{r}}|^3 = 0. \tag{96}$$

$$\nabla \cdot \mathbf{B} = -(4\pi k_e/c^2)q/|\vec{\mathbf{r}}|^3 + (4\pi k_e/c^2)q/|\vec{\mathbf{r}}|^3 = 0 \quad \Rightarrow \quad \mu_0 = (4\pi k_e/c^2), \tag{97}$$

where $\mu_0 = 4\pi k_e/c^2$ conforms to the SI units $kg \cdot m \cdot C^{-2} = kg \cdot m \cdot s^{-2}A^{-2}$.

Note: In the last section of this article, it will be shown that the CODATA definition, $\mu_0 = 2h\alpha/cq_e^2$ [17], and $\mu_0 = 4\pi k_e/c^2$ both reduce to the same equation, $\mu_0 = 4\pi k_m/q_p^2$.

Faraday's law

From the magnetic field equation 94, where the electric and magnetic fields are propagating at the speed, $v = c$:

$$B = (2\pi k_e/c^2)qv/r^2 \quad \wedge \quad v = c \quad \wedge \quad r = ct \quad \Rightarrow \quad B = (2\pi k_e/c^3)q/t^2. \tag{98}$$

$$B = (2\pi k_e/c^3)q/t^2 \quad \Rightarrow \quad \partial B/\partial t = -(4\pi k_e/c^3)q/t^3. \tag{99}$$

$$\partial B/\partial t = -(4\pi k_e/c^3)q/t^3 \quad \wedge \quad r = ct \quad \Rightarrow \quad \partial B/\partial t = -4\pi k_e q/r^3. \tag{100}$$

Faraday's law is the case, where the electric field, $\mathbf{E}$, is a single radial vector, $\mathbf{E}_{\tilde{\mathbf{r}}_1}$ acting on some point, P. Therefore, at point, P, $\mathbf{E}_{\tilde{\mathbf{r}}_2} = 0$ and $\mathbf{E}_{\tilde{\mathbf{r}}_3} = 0$. From the Lorentz equation 95, the magnetic field, $\mathbf{B}$ at point P, is perpendicular to $\mathbf{E}_{\tilde{\mathbf{r}}_1}$ (aligned on $\tilde{\mathbf{r}}_2$). For a constant force, the Lorentz equation makes the electric field, $\mathbf{E}_{\tilde{\mathbf{r}}_1}$, a function of $\mathbf{B}_{\tilde{\mathbf{r}}_2}$. Combining with equation 89:

$$\mathbf{E}_{\tilde{\mathbf{r}}_1} = 2\pi k_e q/|\tilde{\mathbf{r}}_2|^2 = f(\mathbf{B}_{\tilde{\mathbf{r}}_2}) \quad \wedge \quad \mathbf{E}_{\tilde{\mathbf{r}}_2} = 0 \quad \wedge \quad \mathbf{E}_{\tilde{\mathbf{r}}_3} = 0 \quad \Rightarrow \quad \nabla \times \mathbf{E} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial \tilde{\mathbf{r}}_1^2} & \frac{\partial}{\partial \tilde{\mathbf{r}}_2^2} & \frac{\partial}{\partial \tilde{\mathbf{r}}_3^2} \\ \mathbf{E}_{\tilde{\mathbf{r}}_1} & 0 & 0 \end{vmatrix}$$

$$= (\frac{\partial \mathbf{E}_{r3}}{\partial \tilde{\mathbf{r}}_2^2} - \frac{\partial \mathbf{E}_{r2}}{\partial \tilde{\mathbf{r}}_3^2})\hat{\mathbf{i}} + (\frac{\partial \mathbf{E}_{r1}}{\partial \tilde{\mathbf{r}}_3^2} - \frac{\partial \mathbf{E}_{r3}}{\partial \tilde{\mathbf{r}}_1^2})\hat{\mathbf{j}} + (\frac{\partial \mathbf{E}_{r2}}{\partial \tilde{\mathbf{r}}_1^2} - \frac{\partial \mathbf{E}_{r1}}{\partial \tilde{\mathbf{r}}_2^2})\hat{\mathbf{k}}$$

$$= 0\hat{\mathbf{i}} + 0\hat{\mathbf{j}} + (0 - \frac{\partial 2\pi k_e q/|\tilde{\mathbf{r}}_2|^2}{\partial |\tilde{\mathbf{r}}_2|^2})\hat{\mathbf{k}} \quad = \quad 4\pi k_e q/|\vec{\mathbf{r}}_2|^3\hat{\mathbf{k}} \quad \equiv \quad \nabla \times \mathbf{E} = 4\pi k_e q/|\vec{\mathbf{r}}|^3. \quad (101)$$

Combining equations 101 and 100 yields Maxwell-Faraday's law [3]:

$$\nabla \times \mathbf{E} = 4\pi k_e q/|\vec{\mathbf{r}}|^3 \quad \wedge \quad \partial \mathbf{B}/\partial t = -4\pi k_e q/|\vec{\mathbf{r}}|^3 \quad \Rightarrow \quad \nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t. \quad (102)$$

Maxwell's wave equations

From the Gauss's electric field equation 90:

$$\mathbf{E} = 2\pi k_e q/|\vec{\mathbf{r}}|^2 \quad \wedge \quad r = |\vec{\mathbf{r}}| \quad \Rightarrow \quad \nabla^2 \mathbf{E} = 12\pi k_e q/|\vec{\mathbf{r}}|^4. \quad (103)$$

$$\mathbf{E} = 2\pi k_e q/|\vec{\mathbf{r}}|^2 \ \wedge \ r = ct \quad \Rightarrow \quad \mathbf{E} = 2\pi k_e q/(ct)^2 \quad \Rightarrow \quad \partial^2 \mathbf{E}/\partial t^2 = (12\pi k_e/c^2)q/t^4. \quad (104)$$

$$\partial^2 \mathbf{E}/\partial t^2 = (12\pi k_e/c^2)q/t^4 \quad \wedge \quad r = ct \quad \Rightarrow \quad \partial^2 \mathbf{E}/\partial t^2 = 12\pi k_e c^2 q/|\vec{\mathbf{r}}|^4. \quad (105)$$

$$\nabla^2 \mathbf{E} = 12\pi k_e q/|\vec{\mathbf{r}}|^4 \ \wedge \ \partial^2 \mathbf{E}/\partial t^2 = 12\pi k_e c^2 q/|\vec{\mathbf{r}}|^4 \quad \Rightarrow \quad \nabla^2 \mathbf{E} = (1/c^2)\partial^2 \mathbf{E}/\partial t^2. \quad (106)$$

$$\nabla^2 \mathbf{E} = (1/c^2)\partial^2 \mathbf{E}/\partial t^2 \quad \wedge \quad \varepsilon_0 \mu_0 = 1/c^2 \quad \Rightarrow \quad \nabla^2 \mathbf{E} = \varepsilon_0 \mu_0 \partial^2 \mathbf{E}/\partial t^2, \quad (107)$$

which is Maxwell's electric field wave equation. And from equation (100):

$$\partial B/\partial t = -4\pi k_e q/r^3 \quad \wedge \quad r = |\vec{\mathbf{r}}| \quad \Rightarrow \quad \nabla^2 \mathbf{B} = 12\pi k_e q/|\vec{\mathbf{r}}|^4. \quad (108)$$

$$\mathbf{B} = 2\pi k_e q/|\vec{\mathbf{r}}|^2 \ \wedge \ r = ct \ \Rightarrow \quad \mathbf{B} = 2\pi k_e q/(ct)^2 \ \Rightarrow \quad \partial^2 \mathbf{B}/\partial t^2 = (12\pi k_e/c^2)q/t^4. \quad (109)$$

$$\partial^2 \mathbf{B}/\partial t^2 = (12\pi k_e/c^2)q/t^4 \quad \wedge \quad r = ct \quad \Rightarrow \quad \partial^2 \mathbf{B}/\partial t^2 = 12\pi k_e c^2 q/|\vec{\mathbf{r}}|^4. \quad (110)$$

$$\nabla^2 \mathbf{B} = -12\pi k_e q/|\vec{\mathbf{r}}|^4 \ \wedge \ \partial^2 \mathbf{B}/\partial t^2 = -12\pi k_e c^2 q/|\vec{\mathbf{r}}|^4 \quad \Rightarrow \quad \nabla^2 \mathbf{B} = (1/c^2)\partial^2 \mathbf{E}/\partial t^2. \quad (111)$$

$$\nabla^2 \mathbf{B} = (1/c^2)\partial^2 \mathbf{E}/\partial t^2 \quad \wedge \quad \varepsilon_0 \mu_0 = 1/c^2 \quad \Rightarrow \quad \nabla^2 \mathbf{B} = \varepsilon_0 \mu_0 \partial^2 \mathbf{B}/\partial t^2, \quad (112)$$

which is Maxwell's magnetic field wave equation.

The fine structure constant, α

The ratio of two subtypes of force implies ratios of the form:

$$\alpha_\tau = \frac{F_{\tau_1}}{F_{\tau_2}} = \frac{K\tau_1^2/r^2}{K\tau_2^2/r^2} = \frac{\tau_1^2}{\tau_2^2}. \tag{113}$$

For example, where q_e is the elementary (electron) charge ($1.60217663 \cdot 10^{-19}\ C$), and q_p is reduced Planck charge unit, and the fine structure electron coupling constant:

$$\alpha = q_e^2/q_p^2. \tag{114}$$

The gravitational fine structure constant, α_G, is the ratio of two subtypes of Newton's gravity force, which simplifies to:

$$\alpha_G = m_e^2/m_p^2. \tag{115}$$

$\hbar$, h, and Einstein-Planck relation

[3][18] Applying both the direct proportion ratio (126), and inverse proportion ratio (61):

$$m = k_m/r \quad \wedge \quad r = ct \quad \Rightarrow \quad m(ct)^2 = (k_m/r)r^2 = k_m r. \tag{116}$$

$$m(ct)^2 = k_m r \quad \Rightarrow \quad E \stackrel{\text{def}}{=} mc^2 = k_m r/t^2. \tag{117}$$

$$E = mc^2 = k_m r/t^2 \ \wedge \ r/t = c \ \Rightarrow \ E = mc^2 = k_m c/t. \tag{118}$$

The next 4 steps show that $k_m c = \hbar$, the *reduced* Planck constant. From the fine structure equation 115:

$$\alpha = q_e^2/q_p^2 \quad \wedge \quad q_e = 1.60217663 \cdot 10^{-19}\ C \quad \wedge \quad \alpha \approx 7.2973525705 \cdot 10^{-3}$$
$$\Rightarrow \quad q_p = \sqrt{q_e^2/\alpha} \approx 1.87554604 \cdot 10^{-18}\ C. \tag{119}$$

From equations 97 and 119:

$$\mu_0 = 4\pi k_e/c_t^2 = 4\pi c_q^2/c_m = 4\pi(r_p/q_p)^2/(r_p/m_p) = 4\pi m_p r_p/q_p^2 = 4\pi k_m/q_p^2. \tag{120}$$

$$\mu_0 = 4\pi k_m/q_p^2 \ \wedge \ q_p \approx 1.87554604 \cdot 10^{-18}\ C \ \wedge \ \mu_0 \approx 1.25663706127(20) \cdot 10^{-6}\ kg\ m\ C^{-2}$$
$$\Rightarrow \quad k_m = \mu_0 q_p^2/4\pi \approx 3.51767294 \cdot 10^{-43}\ kg\ m. \tag{121}$$

$$k_m \approx 3.51767294 \cdot 10^{-43}\ kg\ m \quad \wedge \quad c = 2.99792458 \cdot 10^8\ m\ s^{-1}$$
$$\Rightarrow \quad k_m c \approx 1.054571817 \cdot 10^{-34}\ kg\ m^2\ s^{-1} = \hbar. \tag{122}$$

Combining equations 118 and 122:

$$E = mc^2 = k_m c/t \; kg \; m^2 \; s^{-2} \;\; \wedge \;\; k_m c = \hbar \; kg \; m^2 \; s^{-1} \;\; \Rightarrow \;\; E = \hbar/t \; kg \; m^2 \; s^{-2}. \tag{123}$$

$$E = \hbar/t \; kg \; m^2 \; s^{-2} \quad \wedge \quad \omega = (2\pi/t) \; radian \; s^{-1} \quad \Rightarrow$$

$$E_\omega = \hbar(2\pi/t) = \hbar\omega \; kg \; m^2 \; radian \; s^{-2}. \tag{124}$$

$$E_\omega = \hbar/t \; kg \; m^2 \; s^{-2} \;\wedge\; f = (1/t) \; cycle \; s^{-1} = ((2\pi/t) \; radian \; s^{-1})/(2\pi \; radian \; cycle^{-1})$$

$$\Rightarrow \quad E_f = \hbar\omega(2\pi/2\pi) = 2\pi\hbar f = hf \; kg \; m^2 \; cycle \; s^{-2} = hf \; kg \; m^2 \; s^{-1} \; Hz. \tag{125}$$

Unit ratio and reduced Planck unit values

The direct proportion ratios are:

$$c_t = c \approx 2.99792458 \cdot 10^8 \; m \; s^{-1}. \tag{126}$$

$$G = c_m c_t^2 \quad \wedge \quad G \approx 6.67430(15) \cdot 10^{-11} \; m^3 \; kg^{-1} \; s^{-2} \quad \Rightarrow$$

$$c_m = G/c_t^2 \approx 7.426162 \cdot 10^{-28} \; m \; kg^{-1}. \tag{127}$$

$$k_e = c_q^2 c_t^2 / c_m \quad \wedge \quad k_e \approx 8.9875517923(13) \cdot 10^9 Nm^2 \; C^{-2} \quad \Rightarrow$$

$$c_q = \sqrt{k_e c_m / c_t^2} \approx 8.6175182 \cdot 10^{-18} m \; C^{-1}. \tag{128}$$

From equation 122, the inverse proportion ratio values are:

$$\hbar \approx 1.054571783 \cdot 10^{-34} \; kg \; m^2/s \quad \wedge \quad c_t = 2.99792458 \cdot 10^8 \; m/s$$

$$\Rightarrow \quad k_m = \hbar/c_t \approx 3.51767283 \cdot 10^{-43} \; kg \; m. \tag{129}$$

$$r_p = \sqrt{r_p^2} = \sqrt{k_m c_m} \approx 1.616255 \cdot 10^{-35} \; m. \tag{130}$$

$$k_t = r_p/c_t \approx 8.71363 \cdot 10^{-79} \; s \; m. \tag{131}$$

$$k_q = r_p/c_q \approx 3.031361 \cdot 10^{-53} \; C \; m. \tag{132}$$

From the fine structure equation 119, the reduced Planck unit values are:

$$q_p = \sqrt{q_e^2/\alpha} \approx 1.87554604 \cdot 10^{-18} \; C. \tag{133}$$

$$r_p = q_p c_q \approx 1.616255 \cdot 10^{-35} \; m. \tag{134}$$

$$t_p = r_p/c_t \approx 5.391246 \cdot 10^{-44} \; s. \tag{135}$$

$$m_p = r_p/c_m \approx 2.176434 \cdot 10^{-8} \; kg. \tag{136}$$

Compton wavelength, λ

[3][18] From equations 63 and 119:

$$\lambda = 2\pi r = 2\pi k_m/m = 2\pi k_m c/mc \quad \wedge \quad h = 2\pi\hbar = 2\pi k_m c \quad \Rightarrow \quad \lambda = h/mc. \tag{137}$$

Schrödenger's wave equation

Start with the previously derived "base" Einstein-Planck relation 123 and multiply the kinetic energy component by mc/mc:

$$mc^2 = \hbar/t \quad \Rightarrow \quad \exists\, V(r,t) : \ \hbar/t = \hbar/2t + V(r,t). \tag{138}$$

$$\hbar/t = h/2t + V(r,t) \quad \Rightarrow \quad \hbar/t = \hbar mc/2mct + V(r,t). \tag{139}$$

And from the distance-to-time (speed of light) ratio (126):

$$\hbar/t = \hbar mc/2mct + V(r,t) \quad \wedge \quad r = ct \quad \Rightarrow \quad \hbar/t = \hbar mc^2/2mcr + V(r,t). \tag{140}$$

$$\hbar/t = \hbar mc^2/2mcr + V(r,t) \ \wedge \ \hbar/t = mc^2 \quad \Rightarrow \quad \hbar/t = \hbar^2/2mcrt + V(r,t). \tag{141}$$

$$\hbar/t = \hbar^2/2mcrt + V(r,t) \quad \wedge \quad r = ct \quad \Rightarrow \quad \hbar/t = \hbar^2/2mr^2 + V(r,t). \tag{142}$$

Multiply both sides of equation 142 by a probability density function, $\Psi(r,t)$.

$$\hbar/t = \hbar^2/2mr^2 + V(r,t) \quad \Rightarrow \quad (\hbar/t)\Psi(r,t) = (\hbar^2/2mr^2)\Psi(r,t) + V(r,t)\Psi(r,t). \tag{143}$$

$$\begin{gathered}(\hbar/t)\Psi(r,t) = ((\hbar)^2/2mr^2)\Psi(r,t) + V(r,t)\Psi(r,t) \quad \wedge \\ \forall\ \Psi(r,t) : \ \partial^2\Psi(r,t)/\partial r^2 = (-1/r^2)\Psi(r,t) \quad \wedge \quad \partial\Psi(r,t)/\partial t = (i\ 1/t)\Psi(r,t) \\ \Rightarrow \quad i\hbar\partial\Psi(r,t)/\partial t = -(\hbar^2/2m)\partial^2\Psi(r,t)/\partial r^2 + V(r,t)\Psi(r,t),\end{gathered} \tag{144}$$

which is the one-dimensional position-space Schrödenger's equation [7][18].

$$\begin{gathered}i\hbar\partial\Psi(r,t)/\partial t = -(\hbar^2/2m)\partial^2\Psi(r,t)/\partial r^2 + V(r,t)\Psi(r,t) \quad \wedge \quad ||\vec{\mathbf{r}}|| = r \\ \Rightarrow \quad \exists\, \vec{\mathbf{r}} : \ i\hbar\partial\Psi(\vec{\mathbf{r}},t)/\partial t = -(\hbar^2/2m)\partial^2\Psi(\vec{\mathbf{r}},t)/\partial\vec{\mathbf{r}}^2 + V(\vec{\mathbf{r}},t)\Psi(\vec{\mathbf{r}},t),\end{gathered} \tag{145}$$

which is the 3-dimensional position-space Schrödenger's equation [7] [18].

Dirac's wave equation

Start with the previously derived "base" Einstein-Planck relation 123:

$$mc^2 = \hbar/t \quad \Rightarrow \quad \exists\, V(r,t):\ mc^2/2 + V(r,t) = \hbar/t \quad \Rightarrow \quad 2\hbar/t - 2V(r,t) = mc^2. \tag{146}$$

$$\forall\, V(r,t):\ V(r,t) = \hbar/t \quad \wedge \quad r = ct \quad \wedge \quad 2\hbar/t - 2V(r,t) = mc^2$$
$$\Rightarrow \quad 2\hbar/t - i2\hbar c/r = mc^2. \tag{147}$$

Use the ratios, $r = c_q q$, and, $r = ct$. to multiply each term on the left side of equation by 1:

$$qc_q/r = qc_q/ct = 1 \wedge 2\hbar/t - i2\hbar c/r = mc^2 \ \Rightarrow\ 2\hbar(qc_q/c)/t^2 - i\hbar((qc_q/c)/r^2)c = mc^2. \tag{148}$$

Applying a quantum amplitude equation in complex form to equation 148:

$$A_0 = 2(c_q/c)((1/t) - i(1/r)) \quad \wedge \quad 2\hbar(qc_q/c)/t^2 - i\hbar((qc_q/c)/r^2)c = mc^2$$
$$\Rightarrow\ \hbar\partial(-qA_0)/\partial t - i\hbar(\partial(-qA_0)/\partial r)c = mc^2. \tag{149}$$

Translating equation 150 to moving (rest frame) coordinates via the Lorentz factor, $\gamma_0 = 1/\sqrt{1-(v/c)^2}$:

$$\hbar\partial(-qA_0)/\partial t - i\hbar(\partial(-qA_0)/\partial r)c = mc^2$$
$$\Rightarrow \quad \gamma_0\hbar\partial(-qA_0)/\partial t - \gamma_0 i\hbar(\partial(-qA_0)/\partial r)c = mc^2. \tag{150}$$

Translating equation 150 to vector form:

$$\gamma_0\hbar(\partial(-qA_0)/\partial t) - \gamma_0 i\hbar(\partial(-qA_0)/\partial r)c = mc^2 \quad \wedge$$
$$||\vec{\mathbf{r}}|| = r \quad \wedge \quad ||\vec{\mathbf{A}}|| = A_0 \quad \wedge \quad ||\vec{\gamma}|| = \gamma_0$$
$$\Leftrightarrow \quad \exists\, \vec{\mathbf{r}}, \vec{\mathbf{A}}, \vec{\gamma} : \gamma_0\hbar(\partial(-qA_0)/\partial t) - \vec{\gamma}\cdot i\hbar(\partial(-q\vec{\mathbf{A}})/\partial r)c = mc^2. \tag{151}$$

Multiplying both sides of equation 151 by the spinor, Ψ, yields:Dirac's equation [8]:

$$\gamma_0\hbar(\partial(-qA_0)/\partial t) - \vec{\gamma}\cdot i\hbar(\partial(-q\vec{\mathbf{A}})/\partial r)c = mc^2 \quad \Rightarrow$$
$$\gamma_0\hbar(\partial(-qA_0)/\partial t)\Psi - \vec{\gamma}\cdot i\hbar(\partial(-q\vec{\mathbf{A}})/\partial r)c\Psi = mc^2\Psi \tag{152}$$

Total of a type

Applying both the direct (126) and inverse proportion ratios (129) to the Euclidean distance:

$$r = \sqrt{r_1^2 + r_2^2}, \quad r_1 = c_\tau \tau, \quad r_2 = k_\tau/\tau, \quad \tau \in \{t, m, q\}$$
$$\Rightarrow \quad r = \sqrt{(c_\tau \tau)^2 + (k_\tau/\tau)^2} \quad \Leftrightarrow \quad \tau = \sqrt{(r/c_\tau)^2 + (k_\tau/r)^2}. \quad (153)$$

Quantum extension to general relativity

The simplest way to demonstrate how to add quantum physics to general relativity is by extending Schwarzschild's gravitational time dilation equation and black hole metric. Apply the total of a type equation 153 to the derivation of Schwarzschild's time dilation and metric (74):

$$\sqrt{1 - (v^2/c^2)} = \sqrt{1 - (v^2/c^2)(r/r)} \quad \wedge \quad r = \sqrt{(c_m m)^2 + (k_m/m)^2} = Q_m$$
$$\Rightarrow \quad \sqrt{1 - (v^2/c^2)} = \sqrt{1 - Q_m v^2/rc^2}. \quad (154)$$
$$\sqrt{1 - (v^2/c^2)} = \sqrt{1 - Q_m v^2/rc^2} \quad \wedge \quad KE = mv^2/2 = mv_{escape}^2$$
$$\Rightarrow \quad \sqrt{1 - (v^2/c^2)} = \sqrt{1 - 2Q_m v_{escape}^2/rc^2}. \quad (155)$$

For a photon, the escape velocity, $v_{escape} = c$.

$$\sqrt{1 - (v^2/c^2)} = \sqrt{1 - 2Q_m v_{escape}^2/rc^2}. \ \wedge \ v_{escape} = c$$
$$\Rightarrow \quad \sqrt{1 - (v^2/c^2)} = \sqrt{1 - 2Q_m c^2/rc^2} = \sqrt{1 - 2Q_m/r}. \quad (156)$$

Combining equation 156 with equation 71 yields Schwarzschild's gravitational time dilation with a quantum mass effect:

$$\sqrt{1 - (v^2/c^2)} = \sqrt{1 - 2Q_m/r} \quad \wedge \quad t' = t\sqrt{1 - (v^2/c^2)} \quad \Rightarrow \quad t' = t\sqrt{1 - 2Q_m/r}. \quad (157)$$

Schwarzschild defined the black hole event horizon radius, $\alpha \stackrel{\text{def}}{=} 2Gm/c^2$. The radius with the quantum extension is $\alpha \stackrel{\text{def}}{=} 2Q_m$. At this point the exact same equations 79 through 82 yield what looks like the same Schwarzschild black hole metric.

Quantum extension to Newton's gravity force

The quantum mass effect is easier to understand in the context Newton's gravity equation than in general relativity, because the metric equations and solutions in the EFEs are much more complex. From equations 153 and 59:

$$m/\sqrt{(r/c_m)^2 + (k_m/r)^2} = 1 \quad \wedge \quad r^2/(ct)^2 = 1 \quad \Rightarrow \quad r^2/(ct)^2 = m/\sqrt{(r/c_m)^2 + (k_m/r)^2}$$
$$\Rightarrow \quad r^2/t^2 = c^2 m/\sqrt{(r/c_m)^2 + (k_m/r)^2}. \tag{158}$$
$$r^2/t^2 = c^2 m/\sqrt{(r/c_m)^2 + (k_m/r)^2} \quad \Rightarrow \quad (m/r)(r^2/t^2) = (m/r)(c^2 m/\sqrt{(r/c_m)^2 + (k_m/r)^2})$$
$$\Rightarrow \quad F \stackrel{\text{def}}{=} mr/t^2 = c^2 m^2/\sqrt{(r^4/c_m^2) + k_m^2}. \tag{159}$$
$$F = c^2 m^2/\sqrt{(r^4/c_m^2) + k_m^2} \quad \wedge \quad \forall\, m \in \mathbb{R},\, \exists\, m_1, m_2 \in \mathbb{R} :\; m_1 m_2 = m^2$$
$$\Rightarrow \quad F = c^2 m_1 m_2/\sqrt{(r^4/c_m^2) + k_m^2}. \tag{160}$$

Quantum extension to Coulomb's charge force

$$q^2/((r/c_q)^2 + (k_q/r)^2) = 1 \quad \wedge \quad r^2/(ct)^2 = 1 \quad \Rightarrow \quad r^2/(ct)^2 = q^2/((r/c_q)^2 + (k_q/r)^2)$$
$$\Rightarrow \quad r^2/t^2 = c^2 q^2/((r/c_q)^2 + (k_q/r)^2). \tag{161}$$
$$(1/r)(r^2/t^2) = (1/r)(c^2 q^2/((r/c_q)^2 + (k_q/r)^2)) \quad \Rightarrow \quad r/t^2 = c^2 q^2/(r^3/c_q^2 + k_q^2/r). \tag{162}$$
$$\forall\, q \in \mathbb{R} :\; \exists\, q_1, q_2 \in \mathbb{R} :\; q_1 q_2 = q^2 \quad \wedge \quad r/t^2 = c^2 q^2/(r^3/c_q^2 + k_q^2/r) \quad \Rightarrow \; \textit{nonumber} \tag{163}$$
$$\exists\, q_1, q_2 \in \mathbb{R} :\; r/t^2 = c^2 q_1 q_2/(r^3/c_q^2 + k_q^2/r). \tag{164}$$
$$r/t^2 = c^2 q_1 q_2/(r^3/c_q^2 + k_q^2/r) \quad \wedge \quad m = r/c_m \quad \Rightarrow \quad F \stackrel{\text{def}}{=} mr/t^2 = (c^2/c_m) q_1 q_2/(r^2/c_q^2 + k_q^2/r^2). \tag{165}$$

INSIGHTS AND IMPLICATIONS

1. The ruler measure (.1) and convergence theorem (.2) were shown to be useful tools for proving that a real-valued equation is an instance of a countable set of n-tuples.

2. The notion of Euclidean distance as the inverse function of a volume, where that volume is the sum of subset Euclidean volumes(.7), suggests defining a distance measure as the inverse function of a volume that is the sum of subset Euclidean and

non-Euclidean volumes,

$$\forall\, n \in \mathbb{N}, \quad f_n, g \in C^2, \quad d = f_n^{-1} v = f_n^{-1}(\textstyle\sum_{i=1}^m f_n(v_i)) = f_n^{-1}(\sum_{i=1}^m g(s_{i,1}, \cdots, s_{i,n})) : \tag{166}$$

(a) shows the intimate relation between distance and volume that definitions, like inner product space and metric space, ignore [9][10];

(b) is a more simple and concise definition of a distance measure, having the metric space properties [9][10].

3. The left side of the distance sum inequality (.10),

$$(\textstyle\sum_{i=1}^m (a_i^n + b_i^n))^{1/n} \leq (\sum_{i=1}^m a_i^n)^{1/n} + (\sum_{i=1}^m b_i^n)^{1/n}, \tag{167}$$

differs from the left side of Minkowski's sum inequality [14]:

$$(\textstyle\sum_{i=1}^m (a_i^n + b_i^n)^{\mathbf{n}})^{1/n} \leq (\sum_{i=1}^m a_i^n)^{1/n} + (\sum_{i=1}^m b_i^n)^{1/n}. \tag{168}$$

(a) The two inequalities are only the same where $n = 1$.

(b) The distance sum inequality (.10) is a more fundamental inequality because the proof does not require the convexity and Hölder's inequality assumptions required to prove the Minkowski sum inequality [14].

(c) $\forall\, n > 1,\ v > 0$: The distance sum inequality term, $v_i^n = a_i^n + b_i^n$: $d = v^{1/n} = (\sum_{i=1}^m v_i^n)^{1/n}$, is the Minkowski distance, which makes it directly related to geometry. But the Minkowski sum inequality term, $d = v^{1/n} = (\sum_{i=1}^m ((v_i^n)^{\mathbf{n}}))^{1/n} = (\sum_{i=1}^m v_i^{\mathbf{n^2}})^{1/n}$, is *not* a Minkowski distance.

(d) The distance sum inequality might be applicable to machine learning.

4. *Combinatorics.* The number of n-tuples, $v_c = \prod_{i=1}^n |x_i|$, was proven to imply the Euclidean volume equation (.4). And the notion of distance was derived as the inverse function of the sum of the number of n-tuples (.7). These equations were derived without relying on the geometric primitives and relations in Euclidean geometry [19][20], axiomatic geometry [21][22][23] [24][25], trigonometry [26] [27] calculus [28][26][29], and vector analysis [5].

5. *Combinatorics.* The commutative property of the operations defining volume and distance equations limits a domain set to $n \leq 3$ elements (.19). Other elements must have different types (are elements of different sets).

   (a) For example, the vector inner product space, in physical space, can only be extended beyond 3 dimensions if and only if the other dimensions have non-distance types, for example, dimensions of time, mass, and charge.

   (b) If there are more than 6 dimensions, $\{r_1, r_2, r_3, t, m, q\}$, then the dimensions are not unordered dimensions $\subseteq \mathbb{R}$. For example, there can be more dimensions $\subseteq \mathbb{R}$, where each dimension has an intrinsic property giving it a position in strict totally ordered set. Or an additional "dimension" might be a set of unordered discrete states or unordered countable values.

   (c) If each type of quantum state is an ISCL, then there are at most 3 states of the same type: 3 orientations per dimension of space (+1,0-1), 3 quark color charges, {red, green, blue}, 3 quark anti-color charges, and so on.

6. A discrete (point) valued state has measure 0 (zero-length interval). The ratio of a time or distance interval length to zero is undefined, which is the reason quantum entangled (discrete) state values exist independent of (non-deterministic with respect to) distance (position), time, mass, and charge. Therefore, a state value in space and time is undetermined until observed, like Schrödenger's poisoned cat being both alive and dead until the box is opened [7].

7. For each unit interval length, r_p, of a 3-dimensional distance, $r \subseteq \mathbb{R}$, there are unit interval lengths of other types of dimensions, forming unit ratios (126): $c_t = r_p/t_p$, $c_m = r_p/m_p$, $c_q = r_p/q_p \Leftrightarrow$ the inverse proportion ratios (61): $k_t = r_p t_p$, $k_m = r_p m_p$, $k_t = r_p q_p$, where r_p, t_p, m_p, and q_p are the reduced Planck units (133).

8. An empirical law *describes* the relations between the variables in an equation. Deriving empirical laws from the ratios *explains* the reasons for relations between the variables in an equation. Further, all the derivations of the physics equations from the ratios are much shorter and simpler than other derivations, which shows that the ratios are an important tool for physicists and engineers.

9. As shown in the subsection deriving the Schwarzschild's gravitational time dilation and black hole metric (74) [15][16] using ratios illustrates a simple way of finding solutions to Einstein's field equations.

10. c_t, c_m, and c_q are the *maximum* ratios: $r = \sqrt{r_1^2 + r_2^2 + r_3^2} \;\Rightarrow\; r_1,\ r_2,\ r_3 \leq r \;\Rightarrow\; \forall\, \tau \in \{t, m, q\} : r_1/\tau,\ r_2/\tau,\ r_3/\tau \leq r/\tau = r_p/\tau_p = c_\tau$. For example, the speeds of gravity and light waves are limited to the speed, c_t.

11. c_t is a component of the constants: $G = c_m c_t^2$, $k_e = c_q^2 c_t^2/c_m$, $\varepsilon_0 = 1/4\pi k_e = 1/4\pi(c_q^2 c_t^2/c_m)$, and $\hbar = k_m c_t$. The only constant, derived in this article, that does not contain c_t is vacuum permeability: $\mu_0 = 4\pi k_e/c_t^2 = 4\pi c_q^2/c_m$.

12. The fine structure electron coupling constant, α was derived from the ratio of two subtypes of charge force that reduces to the unit-less ratio $\alpha = q_e^2/q_p^2 \approx 0.0072973526$ (115), which is the empirical CODATA value [17].

    (a) The CODATA electron coupling version of the fine structure constant, α is defined as: $\alpha = q_e^2/4\pi\varepsilon_0\hbar c) = q_e^2/2\varepsilon_0 hc$ [17]. The following steps show that the CODATA definition reduces to the ratio-derived equation:

$$\begin{aligned} \varepsilon_0 = 1/4\pi k_e = 1/(4\pi(c_q^2 c_t^2/c_m) \quad \wedge \quad \hbar = k_m c_t \quad \wedge \quad h = 2\pi\hbar \\ \Rightarrow \quad \varepsilon_0 hc = 2\pi k_m c_t^2/(4\pi(c_q^2/c_m)c_t^2) = k_m/(2(c_q^2/c_m)) \\ = m_p r_p/(2((r_p/q_p)^2/(r_p/m_p))) = q_p^2/2. \end{aligned} \tag{169}$$

$$\alpha = q_e^2/2\varepsilon_0 hc \quad \wedge \quad \varepsilon_0 hc = q_p^2/2 \quad \Rightarrow \quad \alpha = q_e^2/2(q_p^2/2) = q_e^2/q_p^2. \tag{170}$$

    (b) The fine structure electron coupling constant is the ratio of electron static charge force to the propagating quantum charge (photon/electromagnetic) wave force, caused by a moving charged particle.

    (c) The gravitational fine structure electron coupling constant, α_G, can be expressed as the ratio of two subtypes of Newton's gravity force, where m_e is the rest electron mass and m_p is the reduced Planck mass unit: $\alpha_G = m_e^2/m_p^2$ (115).

    The currently used gravitational fine structure constant reduces to the equation derived in this article: $\alpha_G = Gm_e^2/\hbar c = (c_m c_t^2)m_e^2/(k_m c_t)c_t = c_m m_e^2/k_m = (r_p/m_p)m_e^2/m_p r_p = m_e^2/m_p^2$.

13. The assumption that the old definition, $\mu_0 = 4\pi k_e/c_t^2 = 1/\varepsilon c^2$, and the newer NIST/CODATA definition, $\mu_0 = 2h\alpha/cq_e^2$[17], are different is *incorrect*! The following steps will show that both equations reduce to the same base equation, $\mu_0 = 4\pi k_m/q_p^2$, which shows that the old and newer NIST/CODATA definitions differ in empirical value because of different measurement methods.

    Using the ratios to reduce the old definition of μ_0 derived in this article:

    $$\mu_0 = 4\pi k_e/c_t^2 = 4\pi c_q^2/c_m = 4\pi(r_p/q_p)^2/(r_p/m_p) = 4\pi m_p r_p/q_p^2 = 4\pi k_m/q_p^2. \quad (171)$$

    Using the ratios to reduce the CODATA definition:

    $$h = 2\pi k_m c_t, \;\; \alpha = q_e^q/q_p^2 \;\; \Rightarrow \;\; \mu_0 = 2h\alpha/cq_e^2 = 4\pi k_m c_t(q_e^2/q_p^2)/c_t q_e^2 = 4\pi k_m/q_p^2. \quad (172)$$

14. Empirical and hypothesized laws of physics use an *opaque* constant to make the units in an equation balance. The opacity has led to the *incorrect* assumptions of those constants being fundamental (atomic) constants.

    In this article, some opaque constants are derived directly from (composed of) the ratios: gravity, $G = c_m c_t^2$ (66), charge, $k_e = c_q^2 c_t^2/c_m$ (85), and reduced Planck $\hbar = k_m c_t$ (119). $\varepsilon_0 = 1/4\pi k_e = 1/4\pi(c_q^2 c_t^2/c_m)$ (90) and $\mu_0 = 4\pi k_e/c_t^2 = 4\pi c_q^2/c_m$ (96).

    And the quantum extensions to: Schwarzschild's gravitational time dilation (156), Newton's gravity force (160), and Coulomb's charge force (165), show that, where the quantum effects become measurable, the constants G, k_e, ε_0, and μ_0 no longer exist (are no longer valid).

    Therefore, G, k_e, ε_0, μ_0, and $\hbar$ are *not* fundamental constants.

15. Constants that use notions of temperature and pressure for example, the Boltzmann constant [17], might not possible to define solely in terms of unit ratios.

16. The derivations of: $\nabla \cdot \mathbf{g} = -4\pi G\rho$ from $\mathbf{g} = 2\pi Gm/|\tilde{\mathbf{r}}|^2$ (67), and $\nabla \cdot \mathbf{E} = -\rho/\varepsilon_0$ from $\mathbf{E} = 2\pi k_e q/|\tilde{\mathbf{r}}|^2$ (90), show that the use of mass and charge density, ρ show that:

    (a) Mass and charge density as motivations are unnecessary complexities. Equation 64 used to derive Newton's law and equation 83 used to derive Coulomb's law relate linear acceleration, r/t^2 to mass and charge. It is small step to recognize

Gauss's $\mathbf{g}$, Poisson's $-\nabla\Phi(\tilde{\mathbf{r}}, t)$, and Gauss's $\mathbf{E}$ relate the angular (orbital) acceleration, $2\pi r/t^2$, to mass and charge, where differentiation yields the gravity field and electric field laws.

The derivations are much simpler and shorter using angular (orbital) acceleration, $2\pi r/t^2$. The orbit length, $2\pi r$, corresponds to the Gauss's line integral length of a curve circumscribing a sphere containing the mass or charge.

(b) Mass and charge density obfuscates the pattern, $\nabla \cdot f(x, r) = -4\pi k_x x/|\tilde{\mathbf{r}}|^3$, being derived from the inverse square pattern, $f(x, r) = 2\pi k_x x/|\tilde{\mathbf{r}}|^2$. The constant, G (derived from the inverse square relation (64)), is used in Einstein's constant, $\kappa = 8\pi G/c^4$, and the use of energy density in the stress energy tensor, $T_{\mu,\nu}$, in Einstein's field equations [5] also obfuscate the inverse square pattern.

17. The derivation Lorentz's law here (94) differs from other relativistic derivations, in that equations for the magnetic field, $\mathbf{B}$, and magnetic permeability, μ_0, are derived in the process. The equation, $\mathbf{B} = (2\pi k_e/c^2)vq/|\vec{\mathbf{r}}|^2$, allows a simple derivation of the Faraday's law (102) and the magnetic component of Maxwell's electromagnetic equations (112).

18. Maxwell's derivation of the electric and magnetic field wave equations required: Gauss's electric field law in a vacuum: $\nabla \cdot \mathbf{E} = 0$, Gauss's magnetic field law in a vacuum: $\nabla \cdot \mathbf{B} = 0$, Faraday's Law: $\nabla \times \mathbf{E} = -\partial\mathbf{B}/\partial t$, and Ampere's Law: $\nabla \times \mathbf{B} = \mu_0\varepsilon_0\partial\mathbf{E}/\partial t$[3]. In contrast, the derivations (107) (112), in this article, did not require any of those laws and relied solely on equations derived from the ratios.

19. Einstein's relativity equations: 1) assume the Lorentz transformations, 2) assume the laws of physics are same at each coordinate point, 3) assume the notion of light, and 4) assume that the speed of light is the same at each coordinate point [6][5].

    The derivations, in this article, were made without those assumptions (does not even require the notion of light). The unit ratios are ratios of Euclidean distance to time, mass, and charge. Therefore, the unit ratios and all equations (laws) and constants derived from those ratios must be the same at each Euclidean-like coordinate point.

20. Using the quantum units, r_p and t_p: r_p/t_p^2, suggests a maximum linear acceleration for masses. And $2\pi r_p/t_p^2$ suggests a maximum orbital or rotational acceleration.

21. The Einstein-Planck relations, $E_\omega = \hbar\omega\ kg\ m^2\ radian\ s^{-2}$ (124) and $E_f = hf\ kg\ m^2\ s^{-1}Hz$ (125), were both derived from the base Einstein-Planck relation, $E = \hbar/t\ kg\ m^2\ s^{-2}$ (123). And the Schrödenger's wave equation (145) [7] and Dirac's wave equation (151) [8] were both derived from the base Einstein-Planck relation.

22. The quantum extensions to: Schwarzschild's gravitational time dilation (156), black hole metric (82), Newton's gravity force (160), and Coulomb's charge force (163) make quantifiable predictions:

    (a) For gravity, $\lim_{r\to 0} F = c^2 m_1 m_2/k_m$, and for charge, $\lim_{r\to 0} F = 0$. Finite maximum gravity and charge forces: 1) eliminates the problem of forces going to infinity as $r \to 0$, 2) might allow radioactivity without the need for a weak force, and 3) finite sloped energy well walls might explain quantum tunneling.

    (b) The quantum-extended relativity and classic equations reduce to the non-extended relativity and classic equations and constants, where the distance between masses and charges is sufficiently large or the masses and charges sufficiently large that the quantum effects are not measurable. *Note* that G, k_e, ε_0, μ_0, and κ (Einstein's constant, which contains G) do not exist (are not valid), where the quantum effects becomes measurable.

    (c) The covariant tensor components, in Einstein's field equations, that had the units $1/distance^2$, will now have the more complex units, $1/\sqrt{(distance^4/c_\tau^2) + k_\tau^2}$, $\tau \in \{t, m\}$.

    (d) $1/\sqrt{(distance^4/c_\tau^2) + k_\tau^2}$ implies that as distance $\to 0$, spacetime curvature peaks at the Planck unit distance, r_p (133). The ratio of quantum units, m_p/r_p^3, might indicate a maximum mass density. A finite force (spacetime curvature) and finite mass density would imply that black holes have sizes > 0 (are not singularities). Black hole evaporation might be possible. If there was a "big bang," then it might not have originated from a singularity.

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